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arta med dagskarta

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Image file:  65b Borrhålskarta med dagskarta.jpg
Image size:  10854 × 7100 pixels
Number of GCPs:  16

Source CRS:  EPSG:3152  (ST74, Stockholm 1938)
Target CRS:  EPSG:3006  (SWEREF99 TM)

Transformation:  pyproj (PROJ pipeline)
Pipeline:  proj=pipeline step inv proj=tmerc lat_0=0 lon_0=18.05779 k=0.99999425 x_0=100178.1808

Inter-consistency (affine fit):
  RMS residual:  0.407 m
  Max residual:  1.194 m
  Outlier threshold (2×RMS):  0.814 m

Overall quality:  GOOD
  (1 outlier GCP(s) detected: 12)
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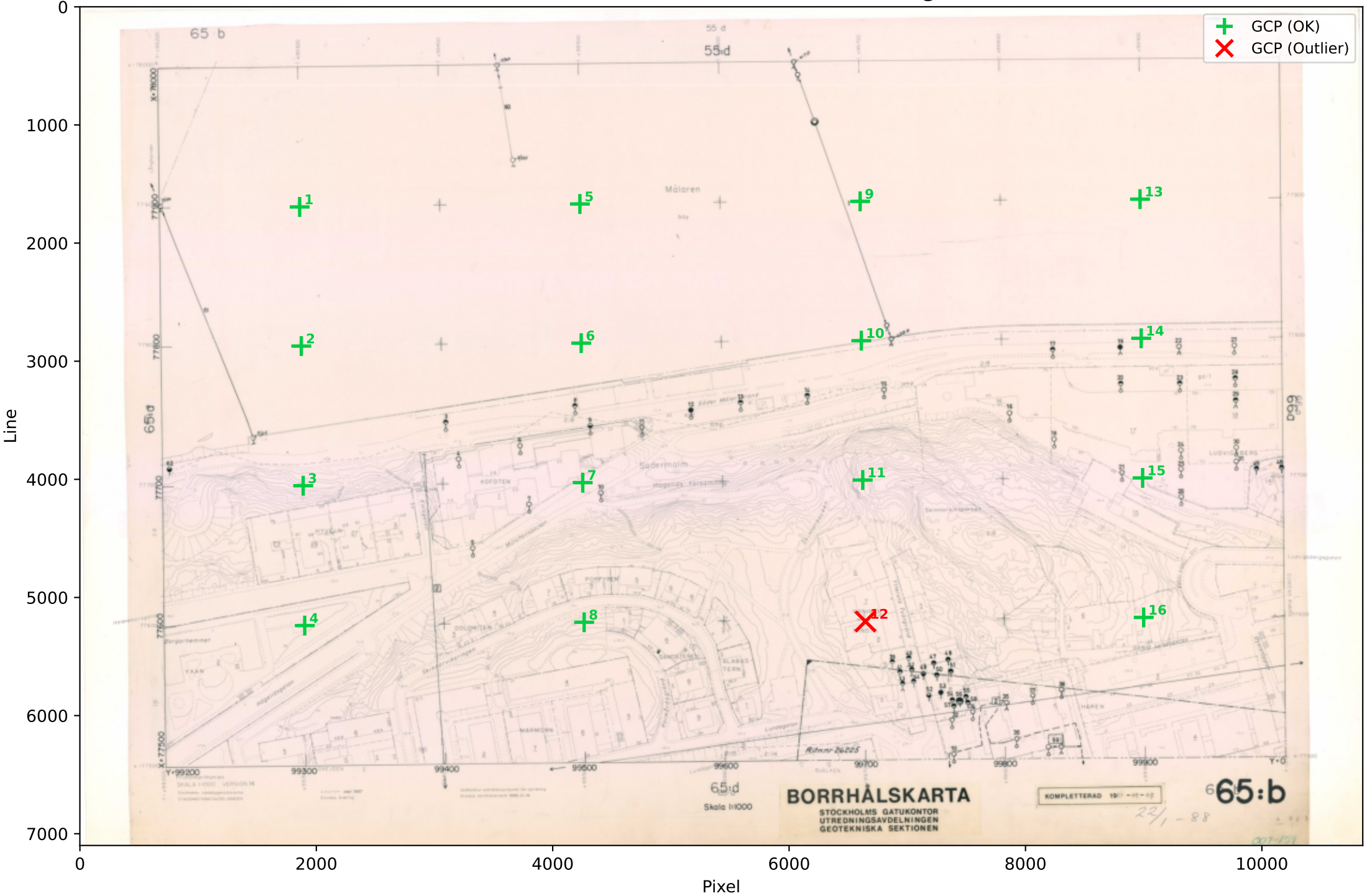
GCP Locations — 65b Borrhålskarta med dagskarta



Inter-consistency Analysis — Affine Fit Residuals

GCP	Pixel	Line	ST74 X	ST74 Y	Res. X (m)	Res. Y (m)	Res. Total (m)	Flag
1	1859	1694	99300	77900	0.194	0.305	0.362	
2	1873	2872	99300	77800	0.103	-0.014	0.104	
3	1889	4054	99300	77700	-0.154	0.007	0.154	
4	1902	5238	99300	77600	-0.156	0.194	0.249	
5	4230	1669	99500	77900	-0.024	0.039	0.045	
6	4241	2847	99500	77800	0.138	-0.283	0.315	
7	4254	4028	99500	77700	0.134	-0.349	0.374	
8	4266	5210	99500	77600	0.214	-0.332	0.395	
9	6601	1647	99700	77900	-0.239	0.025	0.240	
10	6611	2826	99700	77800	0.008	-0.212	0.213	
11	6623	4006	99700	77700	0.087	-0.364	0.374	
12	6646	5203	99700	77600	-0.747	0.931	1.194	OUTLIER
13	8969	1628	99900	77900	-0.198	0.264	0.330	
14	8979	2806	99900	77800	0.048	-0.059	0.076	
15	8991	3988	99900	77700	0.129	-0.041	0.135	
16	9000	5169	99900	77600	0.462	-0.111	0.475	

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Affine model:  map_x = 0.084435·pixel + -0.000926·line + 99144.41
               map_y = -0.000780·pixel + -0.084609·line + 78044.47

RMS residual: 0.407 m    |    Max residual: 1.194 m    |    Outlier threshold (2×RMS): 0.814 m

Grid spacing (map coords):  ΔX = [200.0, 200.0, 200.0]    ΔY = [100.0, 100.0, 100.0]
Pixel spacing (adjacent columns):  mean=2368.0  std=5.7  range=[2354, 2380]
Line spacing (adjacent rows):  mean=1181.8  std=4.9  range=[1178, 1197]
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Coordinate Transformation — EPSG:3152 → EPSG:3006

GCP	ST74 X	ST74 Y	SWEREF99TM E	SWEREF99TM N
1	99300	77900	673116.644	6579875.451
2	99300	77800	673121.233	6579775.559
3	99300	77700	673125.822	6579675.667
4	99300	77600	673130.412	6579575.775
5	99500	77900	673316.428	6579884.630
6	99500	77800	673321.017	6579784.738
7	99500	77700	673325.606	6579684.846
8	99500	77600	673330.196	6579584.954
9	99700	77900	673516.212	6579893.808
10	99700	77800	673520.801	6579793.916
11	99700	77700	673525.391	6579694.024
12	99700	77600	673529.980	6579594.132
13	99900	77900	673715.996	6579902.987
14	99900	77800	673720.586	6579803.095
15	99900	77700	673725.175	6579703.203
16	99900	77600	673729.764	6579603.310

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Source CRS:  EPSG:3152  —  ST74, Stockholm 1938 (local TM)
Target CRS:  EPSG:3006  —  SWEREF99 TM (ETRS89 / UTM zone 33N)

PROJ pipeline:
  proj=pipeline step inv proj=tmerc lat_0=0 lon_0=18.05779 k=0.99999425 x_0=100178.1808 y_0=-6500614.7836 ellps=GRS80 step proj=utm zone=33 ellps=GR

This is equivalent to Lantmäteriet GTRANS.  The pipeline performs:
  1. Inverse ST74 Transverse Mercator projection (→ geographic on GRS80)
  2. Forward UTM zone 33 projection (→ SWEREF99TM)

No datum shift is required — both CRS use GRS80 ellipsoid and
SWEREF99/ETRS89 realization.  Round-trip accuracy: < 0.001 mm.
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